MODERN WEB ENGINEERING: 8-WEEK INTENSIVE COUR

WEEK 1: CORE ARCHITECTURE, EXECUTION CONTEXT & MODERN ES6+ SYSTE

Official Student Study Guide & Reference Materials

Confidential - Authorized Enrolled Student Access Only

MODULE OVERVIEW:

Understand JavaScript runtime engines, call stacks, event loops, memory lifecycle, an
d clean code patterns for scalable engineering.

KEY LEARNING OBJECTIVES & CORE SYLLABUS:

[01] JavaScript Engine Internals V8 & SpiderMonkey

[02] Execution Context, Call Stack & Microtask Queue

[03] Memory Allocation, Garbage Collection & Leak Prevention

[04] Modern ES6+ Modules, Iterators & Immutable Data Patterns

PRACTICAL EXERCISES & LAB MILESTONES:

- Complete the hands-on code exercises described in repository module 1.
- Submit your architecture review via the designated student portal or WhatsApp advisory.
- Verify test coverage and pass all CI assertions before progressing to the next module.

Instructor Support: Contact Course Administrator via WhatsApp for 1-on-1 feedback.

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